CSE400 Lecture Scribe

04/05/2026 22:52:13

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Here is a faithful, structured lecture scribe for CSE400 Lecture 21–22: Inequalities…

L21_L22.pdf
PDF

Generate a lecture scribe intended to serve as exam-reference material for CSE400 Lecture 21–22: Inequalities and Tail Bounds.

The scribe must faithfully reconstruct the lecture content so that a student can reliably revise the lecture using only this document.

Use only the provided context (lecture slides/PDF and relevant textbook excerpts). Do not introduce any content, explanations, examples, interpretations, or intuition that are not explicitly present in the provided context.

While writing the lecture scribe, reason step-by-step through all definitions, notation, assumptions, derivations, interpretations, and worked examples exactly as presented in the lecture. Ensure that all logical dependencies between ideas are explicitly explained so that the reasoning can be followed during exam revision.

The lecture scribe must clearly and systematically reconstruct the lecture structure, including:

1. Motivation for Tail Probabilities
   - Definition of tail probabilities (e.g., $P(X \geq x)$)
   - Explanation of why computing tail probabilities exactly is difficult
   - Real-world examples provided in the lecture (job arrivals, servers, Poisson process, etc.)
   - Need for bounds instead of exact computation

2. Markov's Inequality
   - Definition and formal statement
   - Conditions under which it applies
   - Step-by-step derivation as shown in the lecture
   - Interpretation of the inequality

- Example(s) exactly as presented

3. Chebyshev's Inequality
   - Definition and formal statement
   - Relation with variance and mean
   - Step-by-step derivation
   - Interpretation of deviation bounds
   - Example(s) from the lecture

4. Chernoff Bound
   - Motivation for tighter bounds
   - Definition and formulation
   - Key idea behind exponential bounding
   - Step-by-step reasoning/derivation as presented
   - Application to probability bounds

5. Poisson Tail Bounds
   - Application of bounds to Poisson random variables
   - Step-by-step formulation and reasoning
   - Any examples or interpretations discussed

6. Jensen's Inequality
   - Definition and conditions (convex functions)
   - Formal statement
   - Interpretation
   - Role in bounding expectations

7. Additional Observations / Case Discussions
   - Any system-level interpretation or insights presented in lecture
   - Any comparisons between different bounds

Include:
- Definitions and notation exactly as introduced
- Assumptions and conditions for each inequality
- Step-by-step derivations where present
- Step-by-step worked examples
- Logical transitions between topics

Maintain a formal academic tone suitable for exam preparation.

Do not:
- Add new examples or explanations
- Simplify into tutorial-style content
- Include any material not present in the provided lecture context

Here is a **faithful, structured lecture scribe** for **CSE400 Lecture 21–22: Inequalities and Tail Bounds**, reconstructed strictly from the provided slides/PDF L21_L22.

CSE400 Lecture 21–22: Inequalities and Tail Bounds — Lecture Scribe

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1. Motivation for Tail Probabilities

Definition of Tail

- The **tail of a random variable** X is defined as:

$$P(X > x)$$

- This represents the probability that the random variable takes **large values beyond a threshold**. L21_L22

Why Do We Care About Tails?

From the lecture slides:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer L21_L22

These correspond to **rare but important events**.

Key Idea

- Mean → average behavior (**easy**)

- Tail → rare extreme events (**hard**) L21_L22

Why Are Tail Probabilities Hard to Compute?

- Requires summing many small probabilities L21_L22

Example 1: Load Balancing (Binomial Setting)

- Jobs are distributed among n servers randomly
- Question:

$$P(X \geq k)$$

where X = number of jobs at a particular server

- Distribution:

$$X \sim \text{Binomial}(n, p)$$

- Exact computation:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

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- Interpretation:
  - Most servers get few jobs
  - Occasionally, one server gets many jobs
  - This is a **tail event (rare overload)** L21_L22

Example 2: Poisson Arrivals

- Jobs arrive with rate λ
- Distribution:

$$X \sim \text{Poisson}(\lambda)$$

- Tail probability:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

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- Interpretation:
  - Most hours → moderate arrivals
  - Sometimes → burst of arrivals
  - This is a **tail event (sudden spike)** L21_L22

Conclusion: Need for Bounds

- Exact tail probabilities:
  - Require long summations
  - Are difficult to compute
- Idea:
  - Instead of exact value → compute **upper bounds**
  - Easier to compute
  - Still provides guarantees L21_L22

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2. Markov's Inequality

Key Idea

- Uses only the **mean**
- Large values are rare L21_L22

Formal Statement

For a non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

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Conditions

- $X \geq 0$
- Expectation $E[X]$ exists

Interpretation

- If mean is small → probability of large deviations is small

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Example (from slide)

- Average = 10
- Probability that values ≥ 100 must be small

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3. Chebyshev's Inequality

Definition

- Uses both **mean and variance**

Formal Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Key Relation

- Mean: $\mu = E[X]$
- Variance: measures spread

Interpretation

- Probability of deviating far from mean is bounded

- Larger variance → weaker bound

Derivation

- Built using Markov's inequality applied to:

$$(X - \mu)^2$$

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4. Chernoff Bound

Motivation

- Markov and Chebyshev give **loose bounds**
- Need **tighter bounds**

Key Idea

- Use **exponential transformation**
- Apply Markov's inequality to:

$$e^{tX}$$

General Form

$$P(X \geq a) \leq \inf_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

Interpretation

- Provides exponentially decreasing bounds
- Much tighter for sums of random variables

Application

- Commonly used for:

- Binomial distributions
- Sums of independent variables

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5. Poisson Tail Bounds

Setting

- $X \sim \text{Poisson}(\lambda)$

Exact Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observation

- Exact computation is complex

Approach

- Apply Chernoff-type bounds
- Obtain simpler upper bounds

Interpretation

- Useful for analyzing:
  - Arrival bursts
  - System overload probabilities

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6. Jensen's Inequality

Definition

- Applies to **convex functions**

Formal Statement

For convex function f:

$$f(E[X]) \leq E[f(X)]$$

Condition

- f must be convex

Interpretation

- Expectation inside function ≤ expectation outside

Role

- Used in:
  - Bounding expectations
  - Deriving Chernoff bounds

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7. Running Example (Used Throughout Lecture)

Setup

- Flip a fair coin n times
- Let:

$$X = \text{number of heads}$$

Distribution

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Tail Question

$$P\left(X \geq \frac{3n}{4}\right)$$

Interpretation

- Tail region = unusually many heads
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Purpose

- Compare:
  - Markov
  - Chebyshev
  - Chernoff

on the same example

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8. Additional Observations / System-Level Insights

Comparison of Bounds

- Markov:
  - Uses only mean
  - Weak bound
- Chebyshev:
  - Uses variance
  - Better than Markov
- Chernoff:
  - Uses exponential method

- Tightest bound

System Interpretation

- Tail events correspond to:
  - Server overload
  - Queue delays
  - Traffic bursts

Key Insight

- Exact computation → complex
- Bounds → simple + safe approximation

Final Key Idea of Lecture

- **Approximate safely instead of computing exactly** L21_L22

If you want, I can next:

- convert this into **exam-ready short notes (1–2 pages)**
- or give **derivations of each inequality step-by-step in detail (like viva answers)**